STAFF REVIEW · H446 123 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Spiral.

2. X is removed; Y remains at the front.

3. O(n).

4. B, because a stack is LIFO.

I Do — source worked steps

1. Model answer — SQL describes the required set of matching records (declarative); the loop specifies iteration, testing and append operations (imperative/procedural). Both can produce equivalent output through different expression styles. — Check: link each claim to the scenario and use precise OCR terminology.

2. Find the parent and children — Alert is the base class; EmailAlert : Alert and PopupAlert : Alert are derived classes. Each is a kind of Alert and can be stored in an Alert[] .

3. Same signature, specialised behaviour — Show() is virtual in Alert . Both derived classes use override to supply their own implementation.

4. Trace the output — a.Show() prints Email then Popup . Although a is declared as Alert , the runtime selects the implementation belonging to the actual object. This is polymorphism, not two separate if tests.

We Do — guided CHECK

1. What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.
Work through the method
Use the middle stage of the visual to follow the process and state each intermediate result.
Shared worked answer
Object-oriented design can group vehicle state and operations within objects. A procedural implementation is possible too; justify selection by encapsulation and reuse rather than claiming one paradigm is always best.

2. Predict the output first
Name the base class and two derived classes. In which order are the actual objects stored in the array?
Apply dynamic dispatch
The first object is a Cat , so its overridden Speak() runs. The second is a Dog . No type-checking if statement is needed.
Explain the actual result
Outputs Miaow then Woof . Animal is the declared reference type; each object’s runtime type

determines the overridden method.

You Do — source mark schemes

1. Paradigm is an organising programming approach [2]; imperative how/state changes [2]; declarative required result/relations [2].

2. SQL expresses target set/condition [3]; procedural loop controls traversal and selection explicitly [3].

3. Inheritance reuses/specialises Alert [2]; overriding supplies each derived class’s Show implementation [2]; a call through Alert uses the object’s actual type at runtime [2]; shared base contract supports extension without scattered type-checking branches [2].

Exit Ticket — source answer

1. False. It emphasises the result or relation rather than control flow.